

Lévy Process Models for NEPSE Equity Returns: Calibration and Forward Simulation for SY Panel Nepal Limited (SYPNL)

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Abstract

We model the daily log-returns of SY Panel Nepal Limited (SYPNL), a stock listed on the Nepal Stock Exchange (NEPSE) on December 10, 2025, using three canonical Lévy process specifications: Merton jump-diffusion, the Variance Gamma (VG) process, and an α -stable process. All three arise as special cases of the Lévy–Khintchine decomposition of a general Lévy process into drift, diffusion, and jump components. Parameters are calibrated to the empirical cumulants (and, for the α -stable case, by maximum likelihood) of daily log-returns, and the fitted processes are used to Monte Carlo simulate a 30-trading-day forward price distribution. We find broad qualitative agreement across the three models on the shape of the forecast distribution, while noting that the short trading history and non-stationary listing-rally period impose significant limits on calibration reliability.

1 Introduction

A Lévy process $\{X_t\}_{t \geq 0}$ is a stochastically continuous process with independent, stationary increments. Every such process decomposes, via the Lévy–Khintchine formula, into a deterministic drift, a Brownian (diffusive) component, and a jump component governed by a Lévy measure ν :

$$\mathbb{E} [e^{iuX_t}] = \exp \left\{ t \left[i\mu u - \frac{1}{2}\sigma^2 u^2 + \int_{\mathbb{R} \setminus \{0\}} (e^{iux} - 1 - iux \mathbb{1}_{|x| < 1}) \nu(dx) \right] \right\}. \quad (1)$$

Equity log-returns are widely modeled as Lévy processes because they exhibit two features that geometric Brownian motion (the classical Black–Scholes assumption) cannot capture: (i) excess kurtosis (fat tails, i.e. large moves are more frequent than a Gaussian model predicts), and (ii) skewness. This motivates comparing three standard extensions.

2 Models

2.1 Merton Jump-Diffusion

Brownian motion with drift plus a compound Poisson jump term, jump sizes $Y_i \sim \mathcal{N}(\mu_J, \sigma_J^2)$, arrival rate λ :

$$X_t = \mu t + \sigma W_t + \sum_{i=1}^{N_t} Y_i, \quad N_t \sim \text{Poisson}(\lambda t). \quad (2)$$

This is a *finite-activity* process (finitely many jumps per unit time) and was introduced by Merton (1976) to price options under crash risk.

2.2 Variance Gamma

Brownian motion subordinated (time-changed) by an independent Gamma process $G_t \sim \text{Gamma}(t/\kappa, \kappa)$:

$$X_t = \theta G_t + \sigma W_{G_t}. \quad (3)$$

VG is pure-jump, infinite-activity, and has become a workhorse model in high-frequency equity return modeling since Madan, Carr, and Chang (1998) because its three parameters (σ : volatility, $\nu \equiv \kappa$: kurtosis control, θ : skewness) map cleanly onto observable return-distribution features.

2.3 α -Stable (Lévy Flight)

Characterized by a self-similar characteristic exponent with tail index $\alpha \in (0, 2]$:

$$\psi(u) = i\mu u - |\sigma u|^\alpha [1 - i\beta \text{sgn}(u) \tan(\pi\alpha/2)]. \quad (4)$$

For $\alpha < 2$ the process has infinite variance and power-law tails, $P(|X| > x) \sim x^{-\alpha}$ — the defining signature of a Lévy flight. Mandelbrot’s original 1963 cotton-price study estimated $\alpha \approx 1.7$ for commodity price changes, motivating its continued use in econophysics.

2.4 Cumulant Matching

A convenient property of any Lévy process is that its cumulants scale linearly in t , so per-day cumulants of empirical log-returns can be matched directly against each model’s theoretical per-unit-time cumulants (c_1 = mean, c_2 = variance, $c_3/c_2^{3/2}$ = skewness, c_4/c_2^2 = excess kurtosis). For Merton, this system is under-determined (four moment equations for five parameters), so the jump rate λ is fixed independently via a robust outlier-detection heuristic (returns more than $3\times$ the median absolute deviation from the median are flagged as jumps) before solving for the remaining four parameters. VG’s three parameters are exactly identified by the first three cumulants. The α -stable parameters are instead estimated by maximum likelihood, since the theoretical variance and higher moments are infinite for $\alpha < 2$ and cannot be moment-matched.

3 Data

Daily OHLC price data for SYPNL were collected from Merolagani (<https://merolagani.com>) covering December 10, 2025 (listing date) through July 10, 2026 — 136 trading sessions. Log-returns are computed as $r_t = \log P_t - \log P_{t-1}$.

A visible feature of the data is that the first ~ 25 sessions show the stock repeatedly hitting NEPSE’s daily 10% circuit limit — a common pattern for newly listed shares experiencing initial price discovery, and not indicative of steady-state i.i.d. return behavior. We therefore report calibration on two windows: the *full sample* (all 135 returns) and a *post-stabilization sample* restricted to February 1, 2026 onward (104 returns), which excludes the circuit-limit period. All forecasting results use the post-stabilization sample.

4 Results

4.1 Fitted Parameters (post-stabilization sample)

Empirically, the post-stabilization daily log-returns have mean -0.0034 , variance 7.83×10^{-4} , skewness 0.69 , and excess kurtosis 0.97 — moderate positive skew and fatter-than-Gaussian tails, consistent with the $\alpha \approx 1.8$ (below the Gaussian limit of 2) recovered by the stable fit.

4.2 30-Day Forward Forecast

Each fitted model was used to Monte Carlo simulate 3,000 independent 30-trading-day price paths forward from the last observed price (NPR 1,359 on July 10, 2026). Figure 1 shows the resulting fan

Model	Parameters	Interpretation
Merton JD	$\mu = -0.0053$, $\sigma = 0.0247$, $\lambda = 0.02$, $\mu_J = 0.0935$, $\sigma_J \approx 0$	Daily drift, diffusive vol, jump rate/size (see Limitations)
Variance Gamma	$\theta = -0.0034$, $\sigma = 0.0284$, $\kappa = 1.85$	Skew, vol, kurtosis-control
α -Stable	$\alpha = 1.81$, $\beta = 1.00$, scale= 0.0177	Tail index, skewness, scale

Table 1: Calibrated parameters, daily log-return scale.

charts; Table 2 gives the terminal (day-30) price distribution.

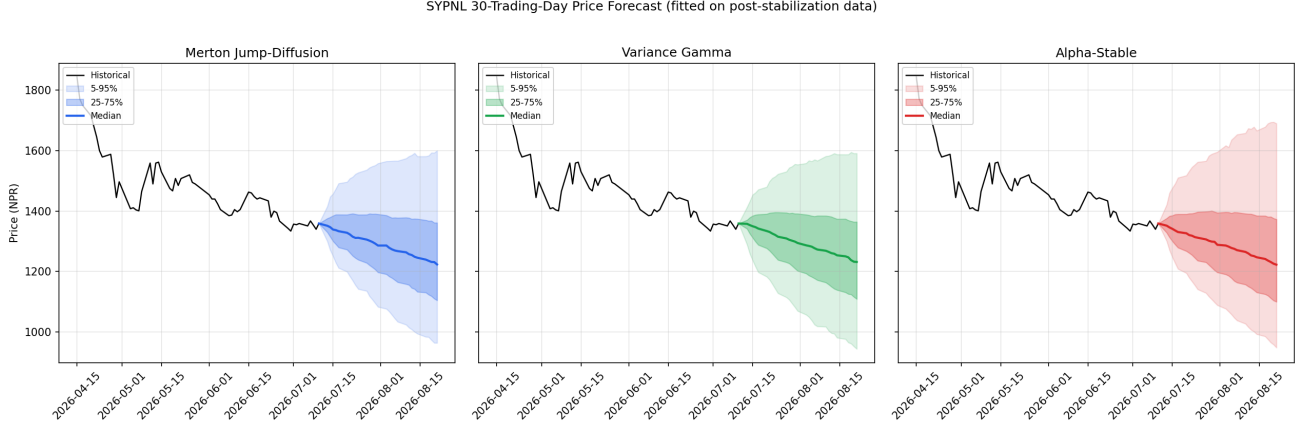


Figure 1: 30-trading-day price forecast fan charts. Shaded bands show the 25–75% and 5–95% ranges of 3,000 simulated paths; solid line is the median path.

Model	p05	p25	Median	p75	p95
Merton JD	964	1,105	1,224	1,362	1,600
Variance Gamma	944	1,109	1,232	1,365	1,591
α -Stable	948	1,100	1,223	1,373	1,690

Table 2: Terminal price distribution (NPR) at $t + 30$ trading days.

All three models agree closely on the central 50% of the distribution (25th–75th percentile within ± 10 NPR of each other). The models diverge most in the upper tail: the α -stable 95th percentile (1,690) exceeds the other two by roughly 90–100 NPR, directly reflecting its heavier tail index. The negative median drift relative to the current price reflects the recent downward trend in the post-stabilization sample (SYPNL fell from a March peak near 2,000 to the 1,300s by July).

5 Limitations

- **Sample size.** 104 post-stabilization observations is small for pinning down third- and fourth-moment (skew/kurtosis) behavior; standard errors on these calibrated parameters are not reported here but are likely large.
- **Merton identifiability.** With only four moment equations available for five parameters, the jump-rate-fixing heuristic is a necessary but somewhat arbitrary simplification; the resulting near-zero σ_J suggests the solver found a degenerate rather than a well-separated jump/diffusion decomposition.

- **i.i.d. assumption.** Lévy processes assume independent, identically distributed increments. SYPNL’s price history (the listing rally in particular) shows visible momentum and mean-reversion structure inconsistent with this assumption.
- **Non-stationarity risk.** As a stock listed less than eight months ago, SYPNL may not yet trade in a stable statistical regime; parameters fitted today may not hold over the 30-day forecast horizon.

6 Conclusion

Despite these caveats, the exercise illustrates a core econophysics point: even with a short, noisy sample, standard Lévy models applied to NEPSE data recover return distributions with visibly fatter tails and non-trivial skew relative to the Gaussian benchmark, and the resulting forecast fan charts communicate the width of realistic price uncertainty far more honestly than a single point forecast would.

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